Polycyclic scheduling
Software pipelining

Same schedule each iteration

Modulo scheduling

Iterative modulo scheduling

Core algorithm

$O(N^3)$

```
II = 1
while (IterativeSchedule(II) != true) do
    II = II + 1;
```

$O(N^2)$

IterativeSchedule() list scheduling, with retry

```
II = MinII
while (IterativeSchedule(II) != true) do
    II = II + 1;
```

ResMII

MinII = Max(resource bound, dependence (recurrence) bound)

RecMII

MII = max(ResMII, RecMII)

Finding ResMII

Simple approach:

Count all FUs used in loop

Divide by number of Ops that use that FU

Max across all FU types

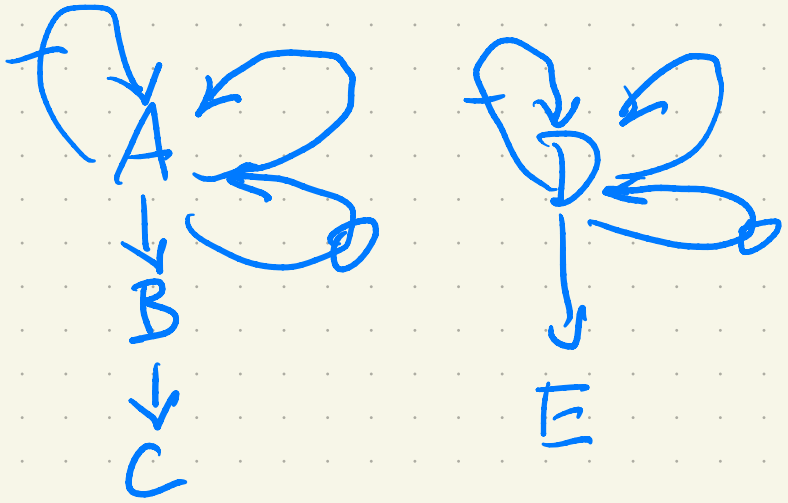
```
Loop: A: Add R2,R2,1
      B: Lw R3,0(R2)
      C: Sw R0,0(R3)
      D: Add R10,R10,1

      Bne R10,0,exit
```

	ALU/branch	Mem
	2	2
# ops:	3	2
	Ceiling $(\frac{3}{2})=2$	Ceiling $(\frac{2}{2})=1$

$ResMII = Max(2, 1) = 2$

```
Loop: A: Add R2,R2,1
      B: Lw  R3,0(R2)
      C: Sw  R0,0(R3)
      D: Add R10,R10,1
```



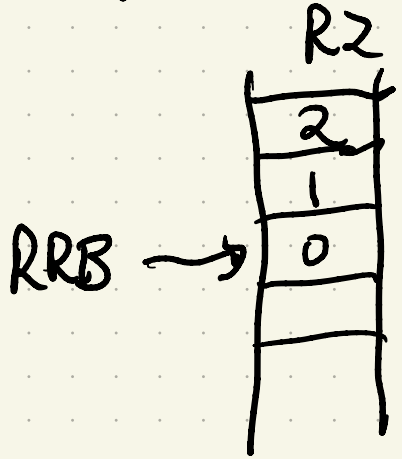
E BrGt-remap R10,0,exit remap means RRB++

Rotating registers: "Register" is now a vector of registers

R2[0], R2[1] ...

R2[RRB]

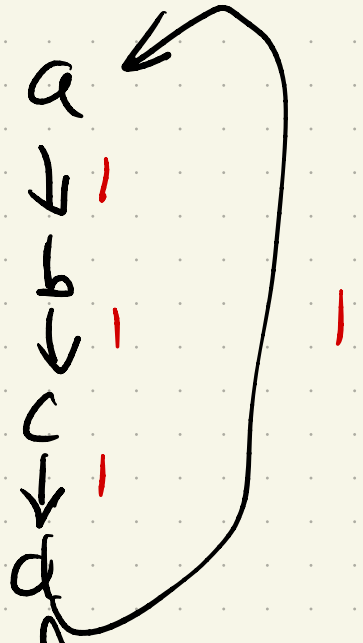
Rotating register base



```
Loop: A: Add R2[0], R2[1], 1
      B: Lw R3[0], 0(R2[0])
      C: Sw R0[0], 0(R3[0])
      D: Add R10[0], R10[1], #1
      E: Brgt-remap R10[0], 0, Loop
```

Finding RecMII

```
a: add r1, r4, ...
b: add r2, r1, -
c: add r3, r2, -
d: add r4, r3, -
```



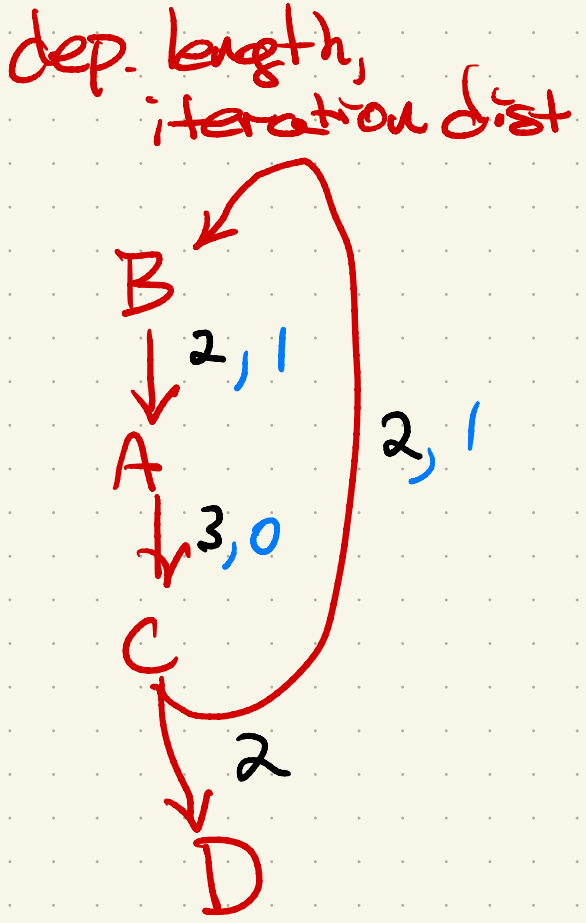
1. Find all cycles in data dep. graph and the cycle's latency = $\sum_{x \in op}$ latencies(x)

2. Find the cycle's iteration distance
   sum of all source reg rotating reg indices

$$(\text{iteration distance}) \times \text{RecMII} \geq \text{cycle length}$$

$$\text{RecMII} = \max_{i \in \text{cycles}} \left(\text{cycle length}_i / \text{iteration dist.}_i \right)$$

```
L: A: Mul R1[0], R3[1], 10
   B: Lw  R3[0], 0(R2[1])
   C: Lw  R2[0], 0(R1[0])
   D: Bne-remap R2[0], 0, L
```



Cycle length = 2+3+2 = 7

Iteration distance = 1+0+1 = 2

$$\text{RecMII} = \text{ceil}\left(\frac{7}{2}\right) = 4$$

Iterative Schedule (II)

blank schedule height (= MultiOps rows) of II

Schedule as before but

if schedule in cycle x in an acyclic schedule,

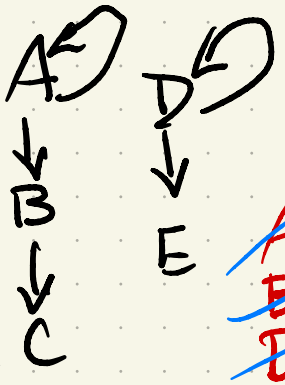
Now schedule in cycle x mod II

→ Attempt all possible schedules if it cannot schedule all ops

ops that have slack

by undoing and trying other options

```
Loop: A: Add R2[0], R2[1], 1
      B: Lw  R3[0], 0(R2[0])
      C: Sw  R5[0], 0(R3[0])
      D: Add R10[0], R10[1], #1
      E: Brgt-remap R10[0], 0, Loop
```



modulo reservation table

	ALU	ALU	MEM	MEM
0	A	D		
1		E	B	C

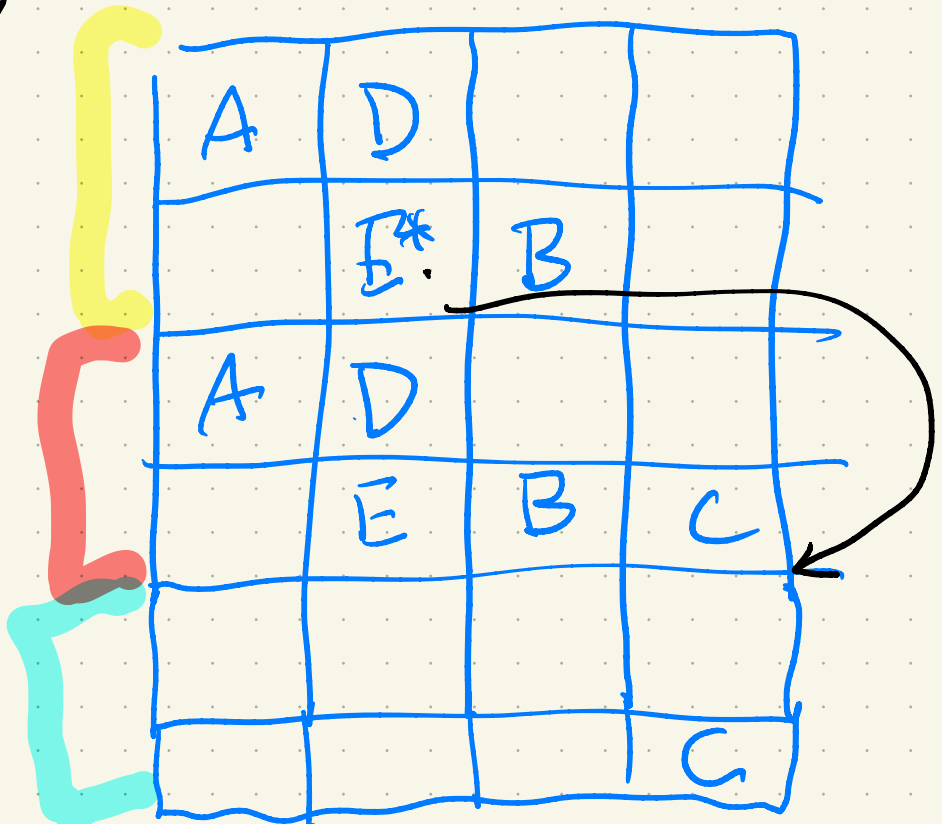
$$MII = \max(RecMII, ResMII)$$

$$= \max(2, 1)$$

$$= 2$$

C cycle 1+2 = 3

3 mod 2 = 1



	Superscalar	VLIW
Pipelined execution	IF ID Disp Sched Ex SU	IF ID EX WB
Renaming	Tomasulo tags or Pregs & RAT	Big register file rotating registers
scheduling resources	sched queue, Phys reg. File, result buses..	None
speculation	Branch prediction and squashing	Global scheduling Predicated execution